Monologue or Dialogue?

Measuring Conversational Structure in the Reasoning Traces of Large Language Models

Oskar Engl

Johannes Kepler University Linz, Austria *oskar.engl@ik.me*

Condensed version — July 2026

Abstract

A reasoning model such as DeepSeek-R1 emits a long "thinking" trace before it answers. That trace can read as one mind reasoning step by step, or as several voices trading proposals and objections. We ask which description the *text* fits when the question is put to measurement. Because no reasoning trace comes labelled with its true voice structure, we build a 2×2 grid of corpora whose structure *is* known — human and AI, monologue and dialogue — and let each detection method judge the traces only after it passes a pre-registered validity gate on the four known corpora. Of three methods, two style-based detectors pass and a dialogue-act detector fails and is set aside. The two survivors *disagree*: a document-level author-count method places the traces just onto the monologue side of its (wide) decision band, while a sentence-level style-switch method places them firmly past the dialogue pole. R1 traces therefore carry local, sentence-level marks of style switching *without* a matching separation into distinct document-level authors — a single narrator whose register shifts, not a transcribed conversation.

1 The question

Reasoning-oriented training is now standard in frontier models: given a problem, a model like DeepSeek-R1 [1] first generates an extended stretch of intermediate tokens [2] — a reasoning trace — and only then commits to an answer. These traces have a curious property. Each is produced by a single model in one pass, yet a trace often contains questions and their answers, objections and concessions, and abrupt reversals ("Wait, that's not right"). In human writing such features can mark a genuine exchange, but they can equally mark one vivid inner voice. Hence a question easy to state and less easy to settle: *does a reasoning trace have the textual structure of a monologue or of a dialogue?*

The question has a live theoretical stake. Kim et al. [3] argue that reasoning-trained models improve partly because they simulate a "society of thought" — distinct internal perspectives that propose, criticise, and reconcile — and older ideas echo it: reasoning as at root argumentative and social [4], and the mind as many interacting agents [5]. Our premise is narrower and testable: voice structure leaves marks in the writing itself. If distinct perspectives take turns producing the text, they should also differ in the small, largely involuntary habits — function-word use, sentence rhythm, dialogue-act moves — that authorship analysis uses to tell writers apart. We therefore treat the question operationally: a monologue should read as one author, a dialogue as several, and we ask which the traces resemble.

2 The design: a calibration grid with a gate

No corpus of reasoning traces is labelled with its true voice structure, which is what makes the question hard to settle directly. We address this with the grid in Figure 1. Four corpora of *known* structure vary producer (human vs. AI) and structure (monologue vs. dialogue): human monologue (think-aloud protocols [6]), human dialogue (group deliberation, DeliData [7]), AI monologue (one Qwen-2.5-7B instance solving a puzzle alone), and AI dialogue (four stateless Qwen-2.5-7B instances, each given a fixed role, debating). The fifth cell — native DeepSeek-R1 traces — is the target.

The grid works as a *gate*. A method may report on the target only after it correctly separates monologue from dialogue on all four known cells; a method that fails is reported as failing and set aside. A detector that cannot tell voices apart in a genuine four-way debate is in no position to rule them out in a trace; one that reports several voices in a lone think-aloud protocol is in no position to affirm them there.

Two design choices keep the contrast clean. First, *task control*: `ai-mono`, `ai-dia`, `ai-reason` and `hum-mono` all use the same puzzle, Game of 24 [8], so a method reading a monologue as one voice cannot merely be reacting to subject matter. (`hum-dia` uses the Wason task, the closest human group-deliberation data available; its role is only to confirm that a method detects human dialogue at all, so it never anchors the placement.) Second, *pre-registration*: every method configuration, gate, threshold and figure was fixed in a written specification and committed to version control (commit `8cbdee4`, 2026-07-17) before any statistic was computed on the traces. Speaker-identity tokens (pseudonyms, role titles, letter labels) were stripped from both dialogue cells beforehand, so dialogue detection cannot be an artefact of speaker cues.

3 Three methods, three signals

The methods read deliberately different signals, so that agreement among them would carry more weight than any one alone.

Method 1 — document-level author count [9]. Clusters a document's paragraph-sized windows on their 50-most-frequent-token profiles and reports the number of clusters as

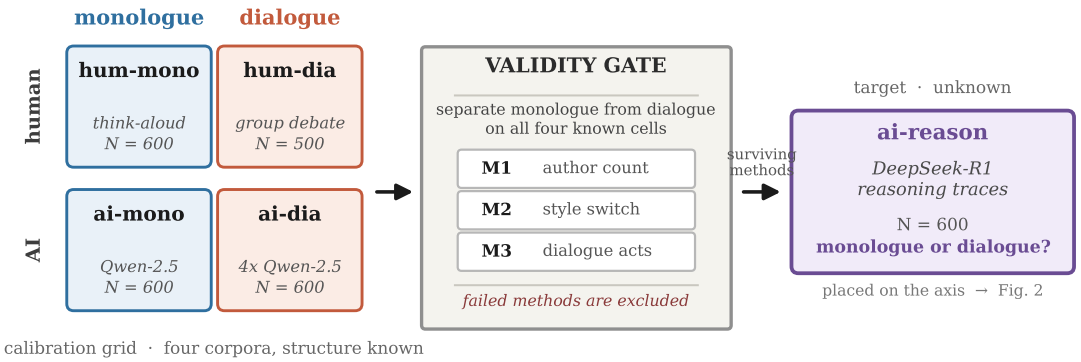


Figure 1: The design. Left to right: four corpora of known voice structure feed a *validity gate* that each of the three detection methods must pass — separating monologue from dialogue on all four known cells — before it is allowed to judge the target. Only surviving methods are applied to the DeepSeek-R1 traces (`ai-reason`) and place them on the monologue–dialogue axis anchored by the two same-task AI cells; the human cells validate only.

an author count, reduced to a binary single/multi call. *Gate:* balanced accuracy across the four cells must beat the 0.50 label-permutation baseline.

Method 2 — sentence-level style switch [10]. A pre-trained sequential sentence-pair classifier emits, for each adjacent sentence pair, the probability that the two were written by different authors, using content-independent style embeddings [11]. Because raw scores depend on length, each document's score is turned into a length-adjusted *lean* — its distance to the dialogue anchor's fitted length curve minus its distance to the monologue anchor's. *Gate:* mean lean must be higher on the dialogue cells than the monologue cells, with Cliff's $\delta \geq 0.33$.

Method 3 — dialogue-act structure [12, 13]. An LLM tags each utterance with a CAMS dialogue act and measures adjacency-pair completion: when a question, request or offer occurs, is it met by a valid answer or acceptance in the very next utterance — the mark of conversation a lone writer has little reason to produce. *Gates:* the completion rate must separate dialogue from monologue cells *and* the tagger's labels must reach Cohen's $\kappa \geq 0.70$ against hand labels.

4 Results: a split

Method 1 passes, but softly. Balanced accuracy is 0.6088 (bootstrap 95% CI $[0.596, 0.621]$; permutation $p = 0.0001$), clearing the baseline. But the pass leans on near-perfect single-author recall (0.975 on `hum-mono`, 0.998 on `ai-mono`) averaged against weak multi-author recall: 0.423 on `ai-dia` and a complete miss on `hum-dia` (0.000). The detector flags dialogue where it finds it but essentially never rules single authorship *out*. A monologue-leaning placement under this method should thus be read as the *absence* of a detected authorship signal, not positive evidence of one voice.

Method 2 passes cleanly. Mean lean is higher on the dialogue cells than the monologue cells (0.0125 vs. −0.0105) with Cliff's $\delta = 0.6211$, well above the 0.33 floor. The traces themselves (0.0433) sit above *both* AI anchors.

Table 1: Cross-method verdicts on the reasoning traces. This is the answer to the research question.

Method	Passed gate?	Verdict on `ai-reason`
1: author count (doc.)	pass	monologue-leaning
2: style switch (sent.)	pass	dialogue-leaning
3: dialogue acts	**fail**	excluded — no reading

Method 3 fails its reliability gate. Its completion rate does separate dialogue from monologue (Cliff's $\delta = 0.4717$), but the tagger reaches only $\kappa = 0.5425$ (95% CI $[0.479, 0.604]$) against hand labels — below the pre-registered 0.70, with the interval not even touching it. Much of the disagreement is between semantically adjacent tags (a tree-weighted α reaches 0.619), but under the frozen rule the method is reported as a failed validation and takes no part in placing the target.

The two survivors disagree (Table 1, Figure 2). Under the placement rule fixed in advance, Method 1 puts the traces just onto the *monologue* side of its own no-decision band ($r = 0.410$ vs. margin $m = 0.361$) — a marginal call the bootstrap reverses in nearly a third of resamples. Method 2 puts them firmly on the *dialogue* side, past the dialogue anchor itself ($r = 0.737$ vs. $m = 0.112$; stable in every resample). By the rule fixed in advance this is reported as a split, not resolved by a vote.

5 What the split means

The claim this study licenses is about *textual surface structure*, not hidden minds: whether the traces exhibit the style-switching and adjacency-pair regularities of multi-party dialogue. Read together, the two survivors describe the traces as text that carries local, sentence-level marks of style switching *without* a matching separation into distinct



Figure 2: The answer in one view. Each surviving method's placement of the target on the monologue–dialogue axis (poles at 0 and 1). The shaded band is that method's margin, within which no lean is called. Method 1's marker sits at the edge of a *wide* margin, just onto the monologue side — a marginal call the bootstrap reverses in nearly a third of resamples. Method 2 lands well past the dialogue pole (stable in every resample). The two calls disagree — the result is a split.

document-level authors. Two qualifications both pull toward the middle: Method 1 is a conservative detector (it recovers multi-voice structure in under a quarter of genuinely multi-voice text), so its monologue call is weak evidence; and Method 2's monologue distance is *extrapolated*, because the traces run far longer than the monologue anchor — what it measures directly is that the traces track the *dialogue* anchor closely across the shared length range.

The upshot is that the outside view and the inside view do not straightforwardly coincide. Kim et al. [3] infer distinct internal perspectives from interpretability and trace labelling; the surface text measured here does not resolve into distinct authorial voices even where its register shifts. Internal multi-agent dynamics can coexist with a stylometrically single-author surface, and the present evidence is consistent with that gap — not with a text that reads, on its face, as a transcribed conversation. We report this as the reading the pre-registered frame supports, not as a refutation of the internal account.

6 Limitations

The target is one model (DeepSeek-R1) on one puzzle family, so the verdict is about this cell, not reasoning traces at large. The target is also a *different* base model than the anchors (Qwen-2.5-7B), so any measured distance carries a shared model offset; the placement is best read as "closer to which pole," not as an absolute distance. Method 2's monologue comparison rests on extrapolation beyond the fitted length range, and all three methods run out of domain. These are the reasons the validity gate exists: a method that survives it has at least shown competence on this data.

References

[1] D. Guo et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* 645:633–638, 2025.

[2] S. Kambhampati et al. Stop anthropomorphizing intermediate tokens as reasoning/thinking traces! *arXiv:2504.09762*, 2025.

[3] J. Kim, S. Lai, N. Scherrer, B. Agüera y Arcas, J. Evans. Reasoning models generate societies of thought. *arXiv:2601.10825*, 2026.

[4] H. Mercier, D. Sperber. *The Enigma of Reason.* Harvard Univ. Press, 2017.

[5] M. Minsky. *The Society of Mind.* Simon & Schuster, 1986.

[6] D. Wurgaft et al. Scaling up the think-aloud method. *Proc. CogSci*, 2025.

[7] G. Karadzhov, T. Stafford, A. Vlachos. DeliData: a dataset for deliberation in multi-party problem solving. *Proc. ACM HCI* 7(CSCW2), 2023.

[8] S. Yao et al. Tree of thoughts: deliberate problem solving with LLMs. *NeurIPS* 36, 2023.

[9] S. Nath. Style change detection by threshold-based and window-merge clustering. *CLEF Working Notes*, 2019.

[10] G. Schmidt et al. Style change detection using a sequential sentence-pair classifier. *CLEF Working Notes*, 2025. *arXiv:2508.00675*.

[11] A. Patel et al. StyleDistance: content-independent style embeddings with synthetic parallel examples. *NAACL*, 2025.

[12] N. Duran, S. Battle, J. Smith. Inter-annotator agreement using the Conversation Analysis Modelling Schema. *Comm. Methods & Measures* 16(3):182–214, 2022.

[13] E. Schegloff, H. Sacks. Opening up closings. *Semiotica* 8(4):289–327, 1973.